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## Quiz 1 – Introduction to OS

L2: Student Name: \_\_\_\_\_

| Q1  | Q2  | Total |
|-----|-----|-------|
| (5) | (5) | (10)  |

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- Q1) In the lab, you developed several programs ranging from “task1.c” to “task 5.c”. Write down one (5)  
Makefile that helps compile all the C codes, and then runs them.

```
CC = gcc

all: task1 task2 task3 task4 task5

task1: task1.c
    $(CC) task1.c -o task1

task2: task2.c
    $(CC) task2.c -o task2

task3: task3.c
    $(CC) task3.c -o task3

task4: task4.c
    $(CC) task4.c -o task4

task5: task5.c
    $(CC) task5.c -o task5

run: all
    ./task1 3 "I hate you"
    ./task2 hello my name is Bilal
    ./task3 11 5
    ./task4 11 5 20 0 9
    ./task5 4 file.txt
```

Q2) Unix-based systems have 6 states: {UNUSED, EMBRYO, SLEEPING, RUNNABLE, RUNNING, ZOMBIE}.

(5)

Briefly inform about the following: {UNUSED, EMBRYO, SLEEPING, RUNNABLE, and ZOMBIE}.

- **UNUSED:** The process table slot is free and not currently allocated to any process.
- **EMBRYO:** The process is in the middle of being created but is not yet ready to run.
- **SLEEPING:** The process is blocked, waiting for an event or resource such as I/O to complete.
- **RUNNABLE:** The process is ready to execute and is waiting in the run queue for CPU time.
- **ZOMBIE:** The process has finished executing, but its parent has not yet called wait() to reap it.